ISA CERTIFIED PERSONAL TRAINER COURSE

WEEK 7

- CHAPTER 10
(MUSCULAR TRAINING: ASSESSMENTS)
- CHAPTER 11
(INTEGRATED EXERCISE PROGRAMMING: FROM EVIDENCE TO PRACTICE)

INTERNATIONAL SPORTS ACADEMY

CHAPTER 10
MUSCULAR TRAINING: ASSESSMENTS

CHAPTER 11
INTEGRATED EXERCISE PROGRAMMING

CONTENT COVERED

Apply the concept of evidence-based practice to exercise programming

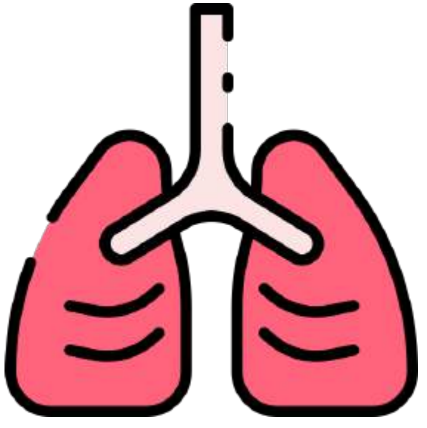
List key considerations for programme maintenance

Identify programming considerations for training recovery

Design personalized, evidence-based Muscular Training programs for clients

BENEFITS OF MUSCULAR TRAINING

[PG 360 – 361]



Increased physical capacity

Enhanced metabolic functions

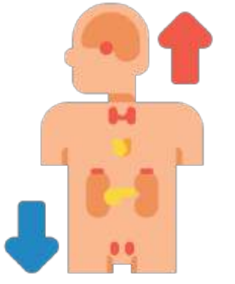
Reduced injury risk and disease prevention

PHYSIOLOGICAL ADAPTATIONS TO ACUTE AND CHRONIC RESISTANCE TRAINING

[PG 365 – 366]

Acute

- ⬆ activation of motor units and muscle fibers
- ⬆ concentrations of catabolic and anabolic hormones



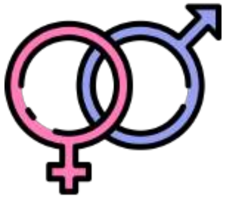
Chronic

- ⬆ strength (neural) and size (hypertrophy)
- Myofibrillar hypertrophy vs Sarcoplasmic hypertrophy

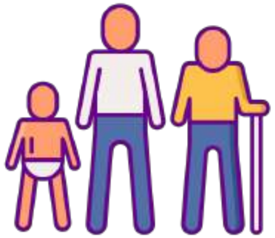
FACTORS THAT INFLUENCE MUSCULAR STRENGTH AND HYPERTROPHY

[PG 366 – 369]

Hormone levels: Anabolic hormones (testosterone and growth hormone)



Sex: Relative strength (M = F)
Absolute strength (M > F)



Age: ↑ age associated with ↓ muscle mass

Muscle fiber type: Type I vs Type II

Muscle Length: Long muscles = greater potential

Limb length: Shorter limb → greater leverage advantages

Tendon insertion point: All else equal, tendon insertion point further allows greater strength production

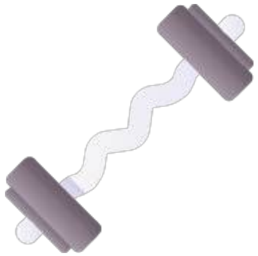
TRAINING PRINCIPLES

[PG 369 – 374]

- **Specificity**
  - Specific to muscles
  - Appropriate protocols for specific sport demands

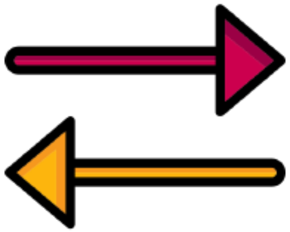
- **Progression**
  - Double-progressive protocol

- **Overload**
  - 5% increments

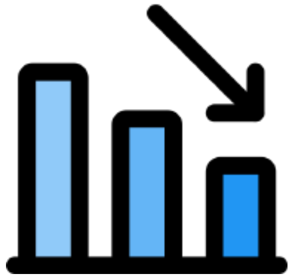


TRAINING PRINCIPLES

[PG 374]



- Reversibility
  - "use it or lose it"



- Diminishing returns
  - Rate of development decrease with training

TRAINING VARIABLES

[PG 375]

- Needs assessment

- Frequency

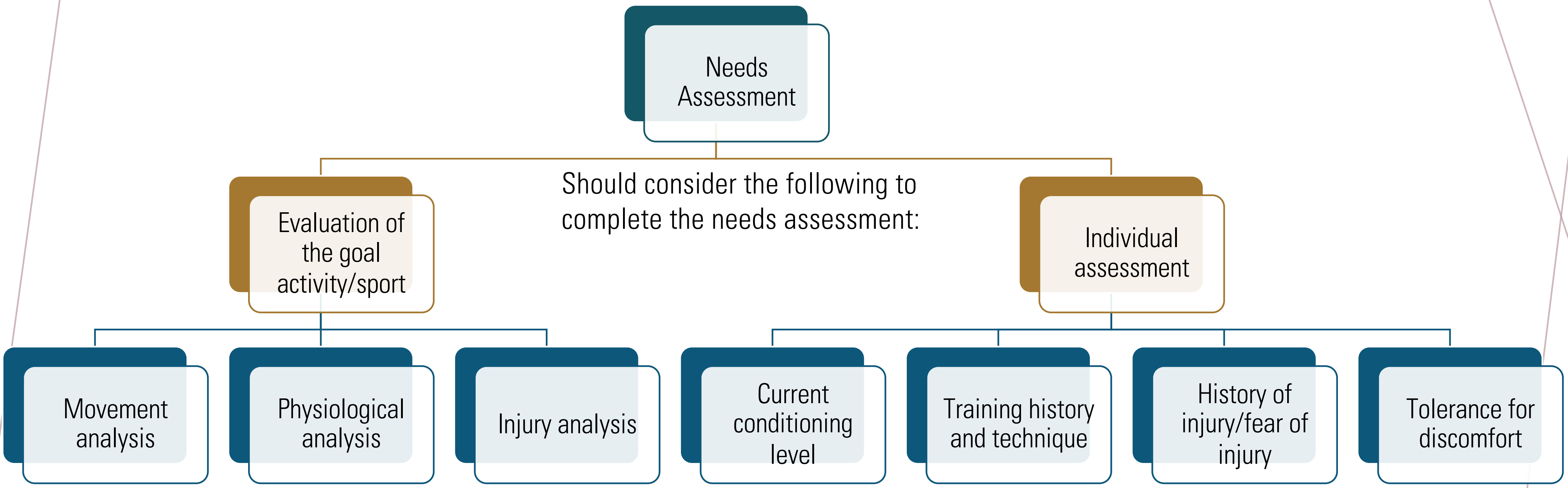
- Exercises selection and order

- Volume and load –
  sets, repetitions, and intensity

- Rest intervals between sets

NEEDS ASSESSMENT

[PG 376]

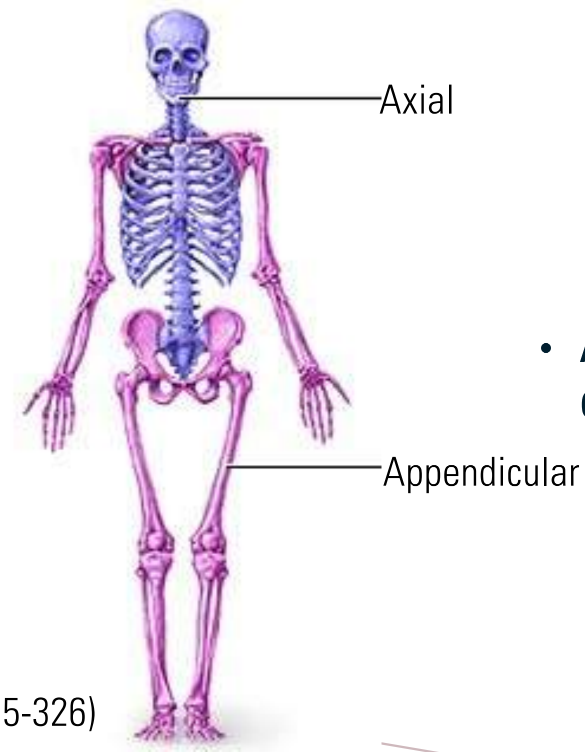


TRAINING FREQUENCY

[PG 377]

- Frequency inversely related to volume and intensity
  - Less vigorous exercises → less muscle microtrauma → require less time for tissue remodelling
- 2-3 days a week
- 48-72 h rest between sessions

TYPES OF RESISTANCE TRAINING EXERCISES



(Pg325-326)

- **Core exercise**: involve two or more primary joints (multi-joint), and engages large muscles while activating synergistic muscles
  - **Structural exercise**: core exercises that load the axial skeleton (places a load on the spine)
  - **Power exercises**: structural exercises that are performed very quickly, e.g. power clean or power snatch
- **Assistance exercise (supplementary exercise)**
  - **Single joint** exercise performed to maintain muscular balance across a joint, prevent injury, or isolate a specific muscle group or muscle

EXERCISE SELECTION AND ORDER

[PG 377 – 378]

- Primary exercises vs assisted exercises

- Variety of methods:
  - Multi-joint followed by single-joint
  - Alternating upper- and lower-extremity exercises
  - Alternating pushing and pulling movements/ targeting agonist and antagonist
  - Supersets vs compound sets
  - Circuit training

TRAINING VOLUME, INTENSITY, REST INTERVALS

[PG 378 – 379]

Volume

- Repetition-volume: Sets x Repetitions
- Load-volume exercise weightload x Sets x Repetitions

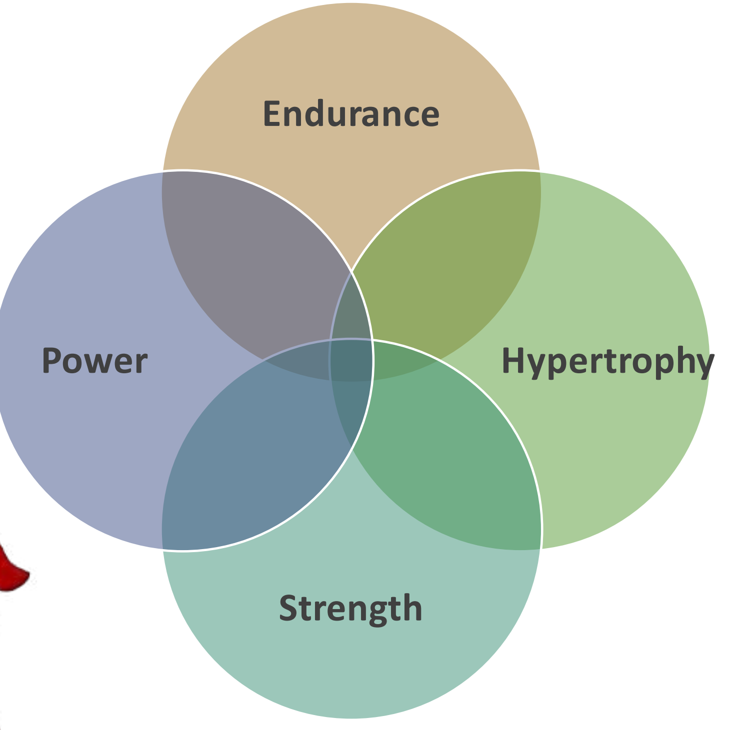
Intensity

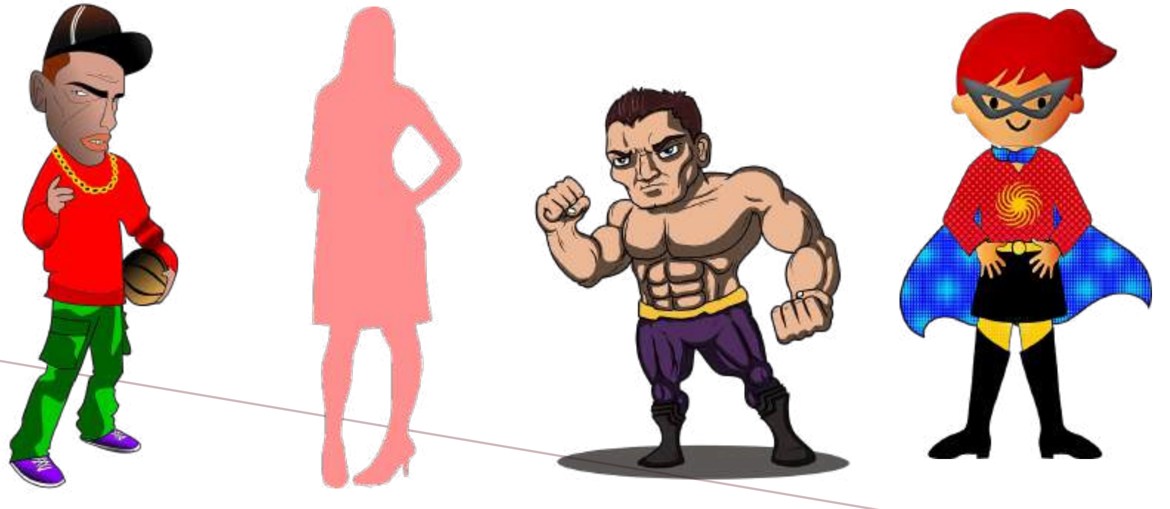
- %1RM
- Perceived effort

Rest Intervals

- Dependent on training goal

PRIMARY RESISTANCE TRAINING GOAL





TRAINING VOLUME, INTENSITY, REST INTERVALS

[PG 379]

Training Goal	Sets	Repetitions	Rest Intervals	Intensity
General muscular fitness	1-4	8-15	2-3 min	20-70% 1RM
Muscular endurance	2-3	≥12	≤30 sec	≤67% 1RM
Muscular hypertrophy	3-6	6-12	30-90 sec	67-85% 1RM
Muscular strength	2-6	≤6	2-5 min	≥85% 1RM
Power: Single-effort events Multiple-effort events	 3-5 3-5	 1-2 3-5	 2-5 min 2-5 min	 80-90% 1RM 75-85% 1RM

Table 9-12: Training Volume Based on Goal

TRAINING TEMPO

[PG 380 – 381]

- Common recommended movement speed
  - 1-3 secs concentric
  - 2-4 secs eccentric
- Controlled movement speeds ranging from 4-8 secs are effective for strength development
- 6-sec repetitions excellent introductory training speeds for new exercisers

COMPONENT OF RESISTANCE TRAINING SESSION

[PG 382 – 383]

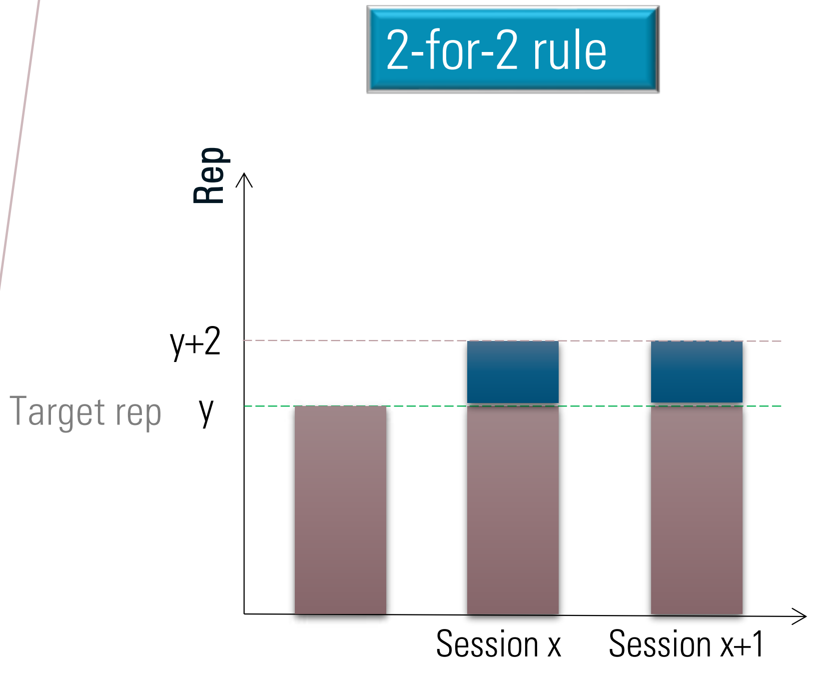
- Warm-up
  - Essential first step of any exercise session
  - Exercises from the ACE IFT Model's Functional and Movement Training phases are sensible choices for use in warming up
- Conditioning
  - A variety of areas of focus in Muscular Training – endurance, strength, power, speed, agility and/or quickness
- Cool-down
  - Exercises from the ACE IFT Model's Functional and Movement Training phases could be good options

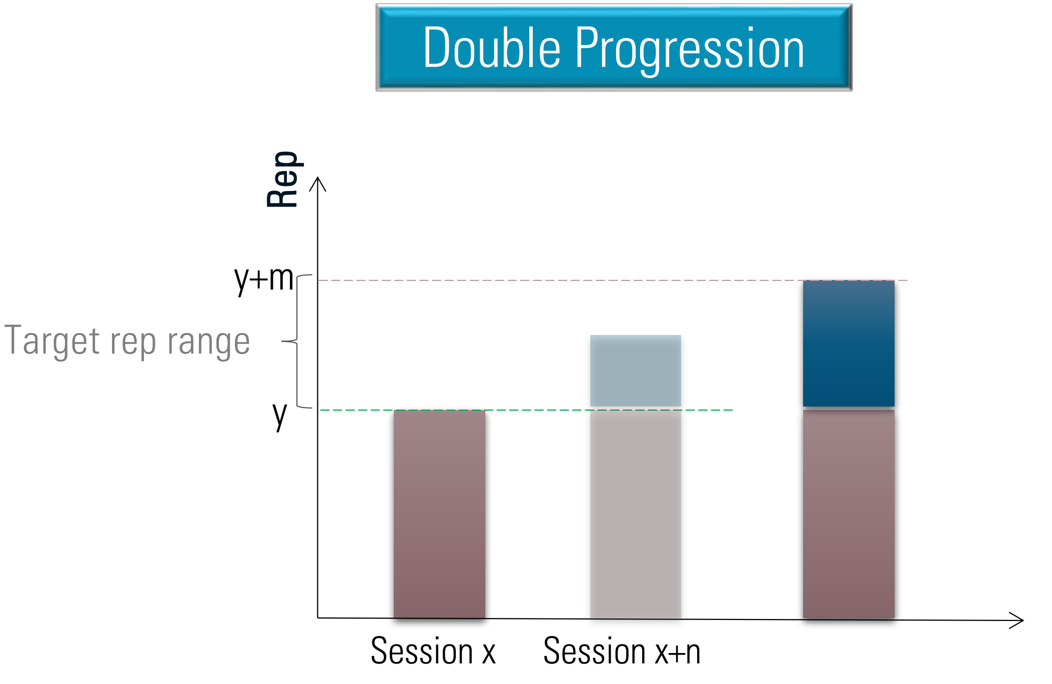
PROGRESSION

- Bodyweight → Machine
- Uniplanar movement machines → Multiplanar movement machines
- Supported machines → Unsupported machines
- Machine weights → Free weights
- Muscle isolation exercises → Multijoint exercises
- Bilateral → Unilateral

PROGRESSION

[PG 370]





ACE-IFT MODEL

[PG 393]

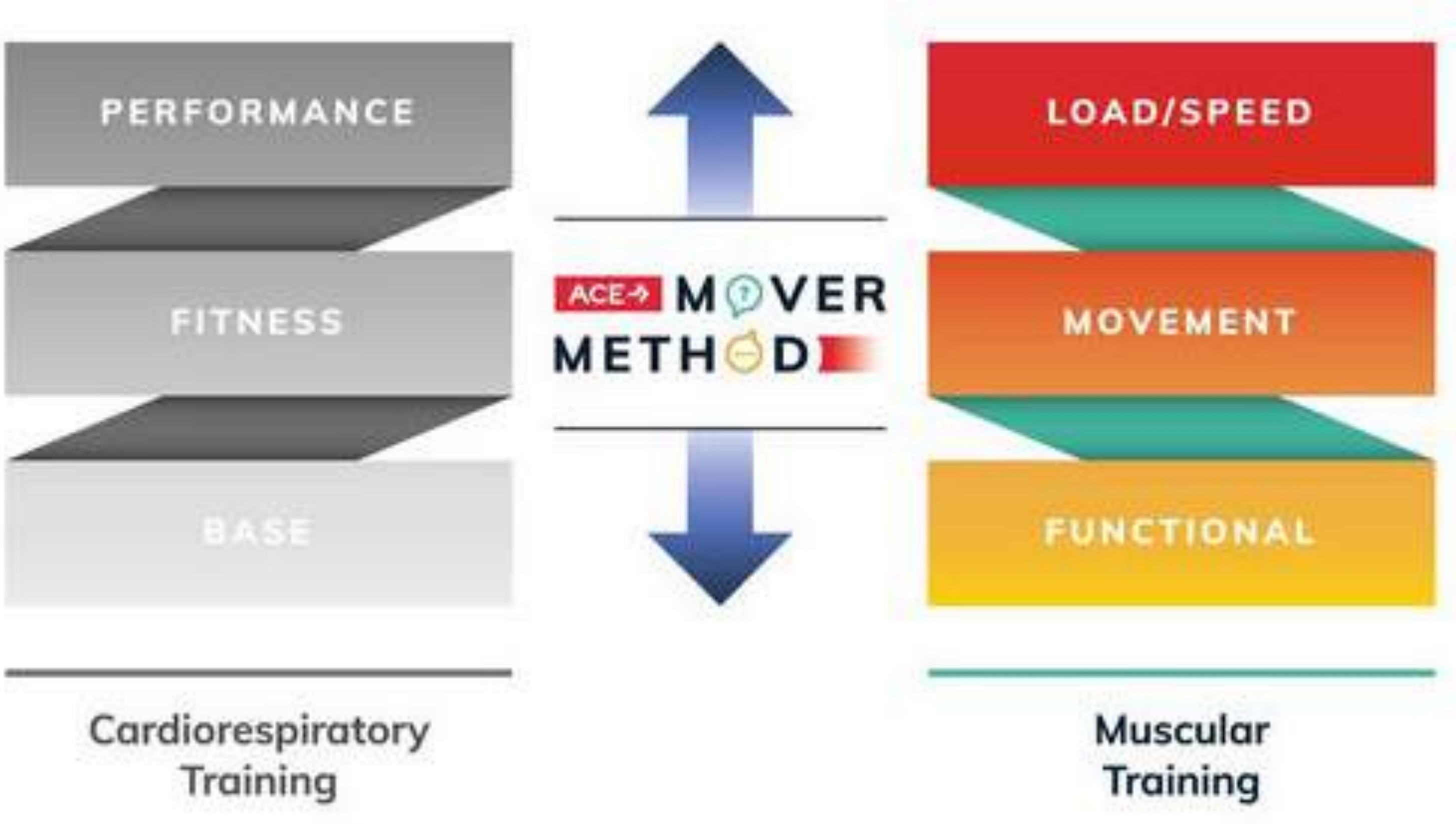


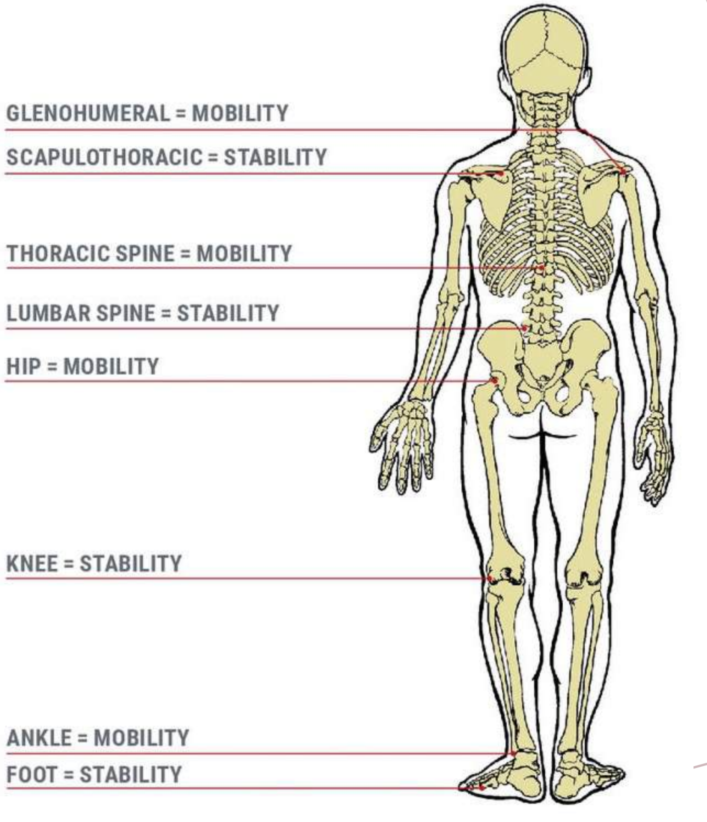
Figure 10-1: ACE IFT Model – Muscular Training

FUNCTIONAL TRAINING

[PG 478]

- Training Focus
  - Establishing/re-establishing postural stability and kinetic chain mobility
  - Core and balance exercises that improves the strength and functions of muscles involved

- Can be incorporated in either warm-up or cool-down as clients progress to movement training and load/speed training



FUNCTIONAL TRAINING

[PG 478 - 479]

- Program design
  - Primarily body weight/body segment weight resistance
  - Promoting stability of the lumbar region then progress to more distal segments

Lumbar (stability)

Hips/thoracic spine (mobility)

Scapulothoracic (stability)

Distal extremities (stability and mobility)

FLEXIBILITY TRAINING

[PG 480 - 484]

- Decreases with age and physical inactivity
- Benefit: Promote increase in joint range of motion
- Commonly used techniques
    - Static stretching
    - Dynamic stretching
    - Ballistic stretching
    - Proprioceptive Neuromuscular Facilitation (PNF)
- Self-myofascial release

Figure 11-13: PNF stretching

Figure 11-14: Self-myofascial release using foam roller

GUIDELINES FOR FLEXIBILITY EXERCISE

[PG 484]

FITT-VP	Evidence-based Recommendation
Frequency	≥2–3 days/week with daily being most effective
Intensity	Stretch to the point of feeling tightness or slight discomfort.
Time	Holding a static stretch for 10–30 seconds is recommended for most adults. In older individuals, holding a stretch for 30–60 seconds may confer greater benefit For proprioceptive neuromuscular facilitation (PNF) stretching, a 3–6 second light-to-moderate contraction (e.g., 20–75% of maximum voluntary contraction) followed by a 10- to 30-second assisted stretch is desirable.
Type	A series of flexibility exercises for each of the major muscle-tendon units is recommended. Static flexibility (i.e., active or passive), dynamic flexibility, ballistic flexibility, and PNF are each effective.
Volume	A reasonable target is to perform 60 seconds of total stretching time for each flexibility exercise.
Pattern	Repetition of each flexibility exercise 2–4 times is recommended. Flexibility exercise is most effective when the muscle is warmed through light-to-moderate aerobic activity or passively through external methods such as moist heat packs or hot baths.
Progression	Methods for optimal progression are not known.

Table 11-7: Flexibility Exercise Evidence-based Recommendations

BALANCE/NEUROMOTOR TRAINING

[PG 486-488]

- Balance exercise: promote static and dynamic balance (e.g., neuromotor exercise)
- Progression: seated → standing → In motion
- Modification of task:
  - Arm placement (support, hands on thighs, hands folded)
  - Surface (chair, balance disk, foam pad, stability ball)
  - Visual (open eyes, sunglasses/dim room, closed eyes)
  - Task (single, multi)
- Refer to Figure 11-15: Sample Progression in Pg487-488 of EPG textbook

BALANCE/NEUROMOTOR TRAINING

[PG 487]

FITT-VP	Evidence-based Recommendation
Frequency	≥2-3 days/week is recommended.
Intensity	An effective intensity of neuromotor exercise has not been determined.
Time	≥20–30 minutes/day may be needed.
Type	Exercises involving motor skills (e.g., balance, agility, coordination, and gait), proprioceptive exercise training, and multifaceted activities (e.g., tai chi and yoga) are recommended for older individuals to improve and maintain physical function and reduce falls in those at risk for falling. The effectiveness of neuromotor exercise training in younger and middle-aged individuals has not been established, but there is probable benefit.
Volume	The optimal volume (e.g., number of repetitions, intensity) is not known.
Pattern	The optimal pattern of performing neuromotor exercise is not known.
Progression	Methods for optimal progression are not known.

Table 11-8: Neuromotor Exercise Evidence-based Recommendations

MOVEMENT TRAINING

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[PG 489]

- Training Focus
  - Developing movement efficiency (five primary movement patterns)
  - Exercise repetition emphasised over exercise intensity

- Program design
  - Assess primary movement patterns
  - Pushing, pulling, squatting movements to be performed linearly and rotationally
  - Add light external resistance (50-60% 1RM)

GUIDELINES FOR RESISTANCE EXERCISE

[PG 490]

FITT-VP	Evidence-based Recommendation
Frequency	Each major muscle group should be trained on 2–3 days/week.
Intensity	60–70% 1-RM (moderate-to-vigorous intensity) for novice to intermediate exercisers to improve strength Experienced strength trainers can gradually increase to ≥80% 1-RM (vigorous-to-very vigorous intensity) to improve strength. 40–50% 1-RM (very light-to-light intensity) for older individuals beginning exercise to improve strength 40–50% 1-RM (very light-to-light intensity) may be beneficial for improving strength in sedentary individuals beginning a resistance-training program <50% 1-RM (light-to-moderate intensity) to improve muscular endurance 20–50% 1-RM in older adults to improve power
Time	No specific duration of training has been identified for effectiveness.
Type	Resistance exercises involving each major muscle group are recommended. Multijoint exercises affecting more than one muscle group and targeting agonist and antagonist muscle groups are recommended for all adults. Single-joint exercises targeting major muscle groups may also be included in a resistance-training program, typically after performing multijoint exercise(s) for that particular muscle group. A variety of exercise equipment and/or body weight can be used to perform these exercises.

Table 11-10 Resistance Exercise Evidence-based Recommendations

GUIDELINES FOR RESISTANCE EXERCISE

[PG 490]

FITT-VP	Evidence-based Recommendation
Type	Resistance exercises involving each major muscle group are recommended. Multijoint exercises affecting more than one muscle group and targeting agonist and antagonist muscle groups are recommended for all adults. Single-joint exercises targeting major muscle groups may also be included in a resistance-training program, typically after performing multijoint exercise(s) for that particular muscle group. A variety of exercise equipment and/or body weight can be used to perform these exercises.
Repetitions	8–12 repetitions are recommended to improve strength and power in most adults. 10–15 repetitions are effective in improving strength in middle-aged and older individuals starting exercise. 15–25 repetitions are recommended to improve muscular endurance.

Table 11-10 Resistance Exercise Evidence-based Recommendations

GUIDELINES FOR RESISTANCE EXERCISE

INTERNATIONAL SPORTS ACADEMY

[PG 490]

FITT-VP	Evidence-based Recommendation
Sets	2–4 sets are recommended for most adults to improve strength and power. A single set of resistance exercise can be effective, especially among older and novice exercisers. ≤2 sets are effective in improving muscular endurance.
Pattern	Rest intervals of 2–3 minutes between each set of repetitions are effective. A rest of ≥48 hours between sessions for any single muscle group is recommended.
Progression	A gradual progression of greater resistance, and/or more repetitions per set, and/or increasing frequency is recommended.

Table 11-10 Resistance Exercise Evidence-based Recommendations

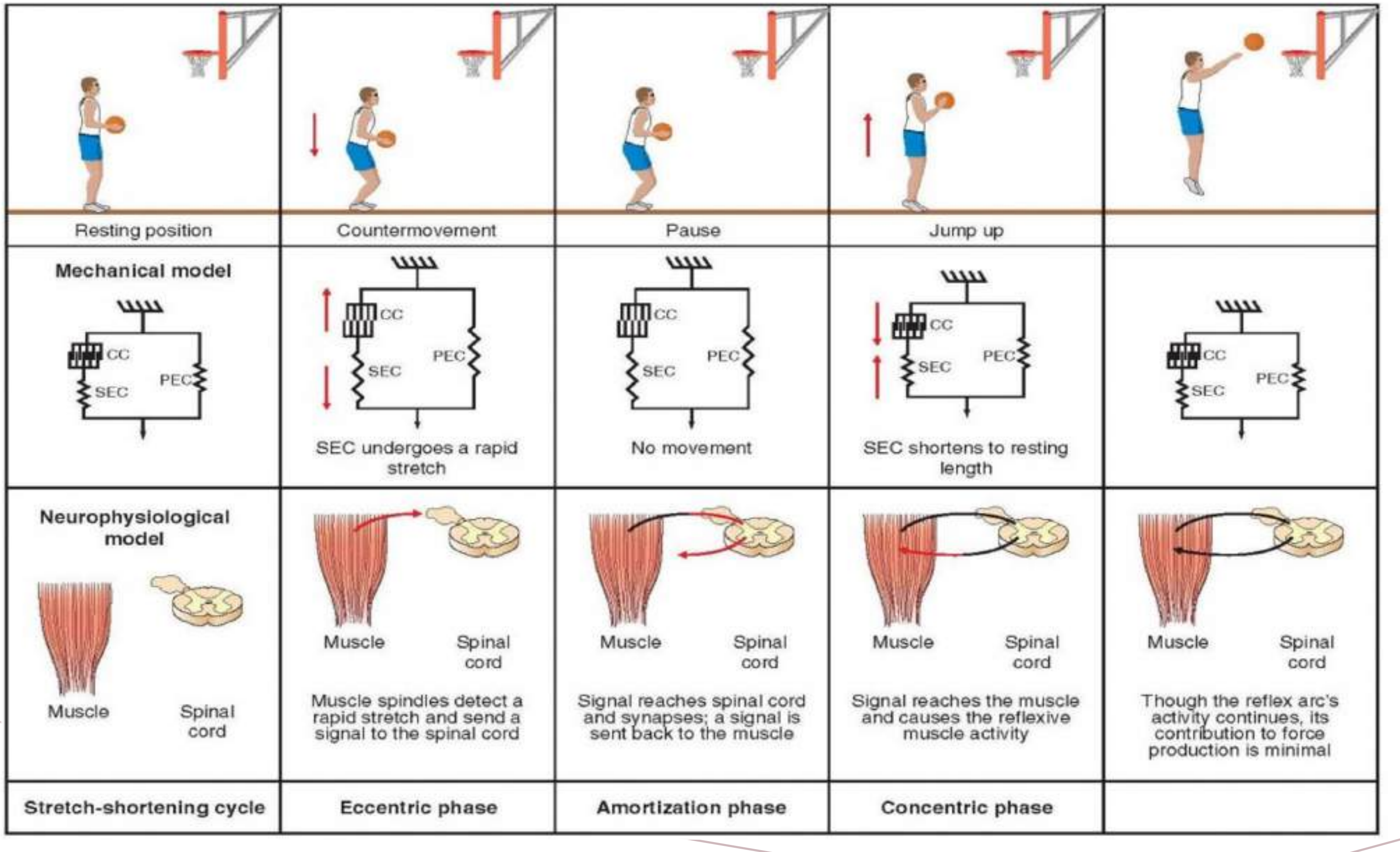
LOAD/SPEED TRAINING

[PG 491 – 492]

- Focus
  - Applying external loads to increase force and/or speed production
  - Muscular endurance, strength, hypertrophy (as per muscular training program design)
  - Performance enhancement (power/speed)
- Program design
  - Assessment to determine strength, power, speed, agility, and quickness
  - Refer to previous guidelines for resistance exercise

PLYOMETRICS AND STRETCH-SHORTENING CYCLE

[PG 494]



Source: Essentials of Strength and Conditioning (National Strength & Conditioning Association), Pg 413, Figure 17.1

PLYOMETRICS

[PG 494 - 495]

- Upper body vs lower body plyometrics
  - Lower body: appropriate for training for any sport
  - Upper body: appropriate for training upper body power for sports such as basketball, volleyball, tennis, softball, rugby etc

- Movement pattern progression for plyometrics

Figure 11-16: Movement-pattern progressions for velocity training

PLYOMETRICS

[PG 495 – 496]

- Precautionary Guidelines
  - Drills performed at the beginning of
    the session after warm-ups
  - Technique is critical
  - Clients should be taught on appropriate landing techniques

PLYOMETRICS

[PG 496 - 497]

- Program design
  - Frequency: 1-3x/week, 48-72 h recovery between sessions
  - Intensity: manipulate factors to modify intensity

Points of contact	The feet are the points of contact for lower-body drills. Single-leg drills impart more stress on the body than double-leg drills.
Speed	Faster movements increase intensity more than slower movements.
Vertical height	The higher the body's center of gravity, the greater the forces of impact upon landing.
Body weight	The greater the client's weight, the more intense the drill. Additional external weight (e.g., medicine ball or weighted vest) can be added to increase the drill's intensity.
Complexity of the exercise	Increasing the complexity, such as adding more body segments or increasing the balance challenge, increases the intensity of the drill.

Table 11-12: Intensity Factors related to Lower-body Plyometric Drills

PLYOMETRICS

[PG 497]

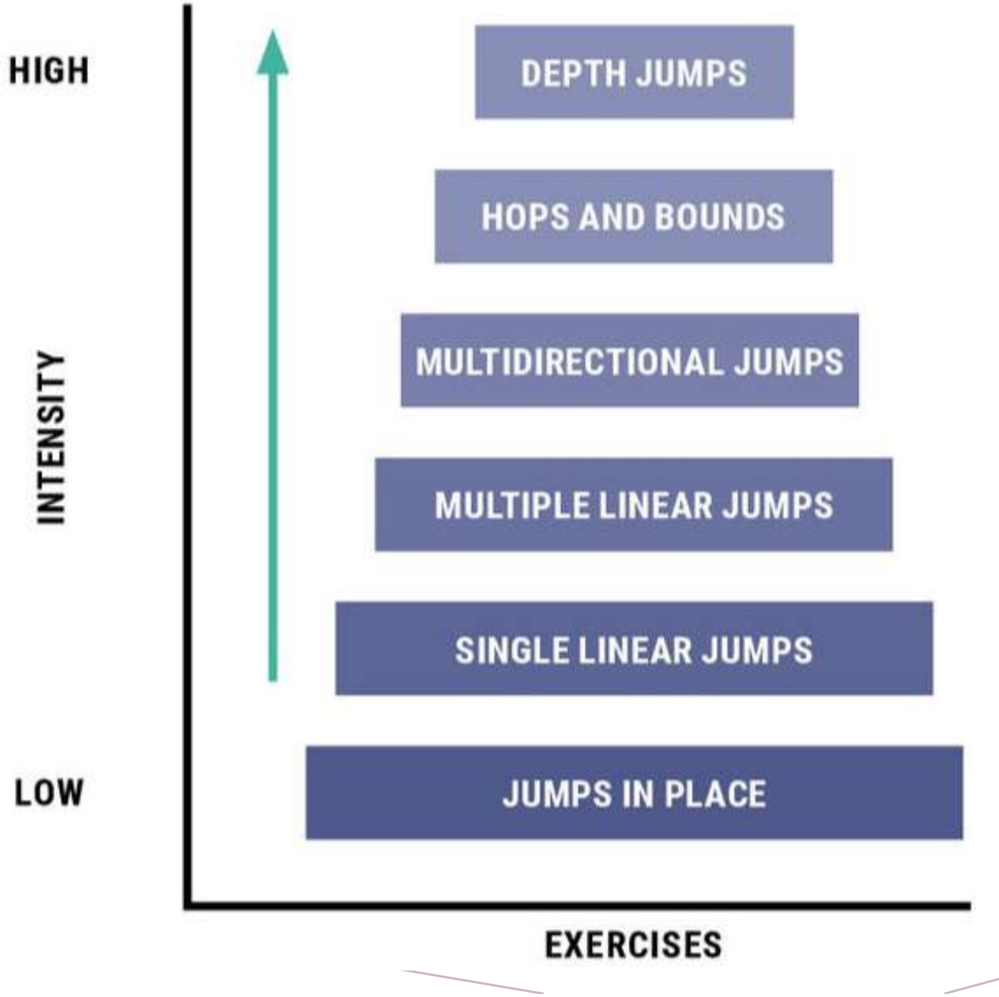


Figure 11-17: Plyometric Drill Classification Model

PLYOMETRICS

[PG 498]

- Repetitions and Sets:

PLYOMETRIC EXPERIENCE	BEGINNING VOLUME
Beginner (no experience)	80-100
Intermediate (some experience)	100-120
Advanced (considerable experience)	120-140

Table 11-13: Plyometric Volume Guidelines (Given in Contacts per Session)

- Type: varied; quick-power movements for upper and lower limbs

SPEED, AGILITY, AND REACTIVITY

[PG 501]

- Speed-strength vs speed-endurance
  - Speed-strength: Ability to develop force at high velocities and relies on a person's reactive ability
  - Speed-endurance: Ability to maintain maximal velocity

- Agility
  - Involves acceleration, deceleration, balance (while changing body position), and reactivity

SPEED, AGILITY, AND REACTIVITY

[PG 502, 516]

- Program design
  - Frequency: 1-3x/week, non-consecutive sessions
  - Intensity:

	Duration	Energy System
Beginner	15–30 seconds	Glycolytic system
Intermediate	<10 seconds	Phosphagen system
Advanced	10–60 seconds	Glycolytic and phosphagen systems

Table 11-16: Duration, Energy System, and Associated Participant Skill Level for Speed and Agility Drills

SPEED, AGILITY, AND REACTIVITY

INTERNATIONAL SPORTS ACADEMY

[PG 502]

- Repetitions and Sets: Volume determined by the duration of time spent working in each of the energy systems

	Sets x Reps	Note
Stationary drills	1-3 sets, 10-15 sec per rep, progressing to 20-30 sec	Training sessions should include 2-3 min rest between repetitions to allow recovery
Dynamic drills	1-3 sets, 20-30 yards per rep, progressing to 100 yards	

- Type: Varied

PERIODIZATION

[PG 507 – 508]

- Systematic process of planned variations in a resistance training program over a training cycle
- Primary goals of periodization are met by appropriately manipulating volume and intensity and by effectively selecting exercises
- Primary advantages of this approach is the reduced risk of overtraining due to the purposeful time devoted to physical and mental recovery
- Typically only core exercises (major muscles, multi joint) are periodized, but all exercises can be varied for intensity and volume

PERIODISATION

[PG 507 – 508]

- Planned progression of exercise that varies the training stimuli (intensity and volume)

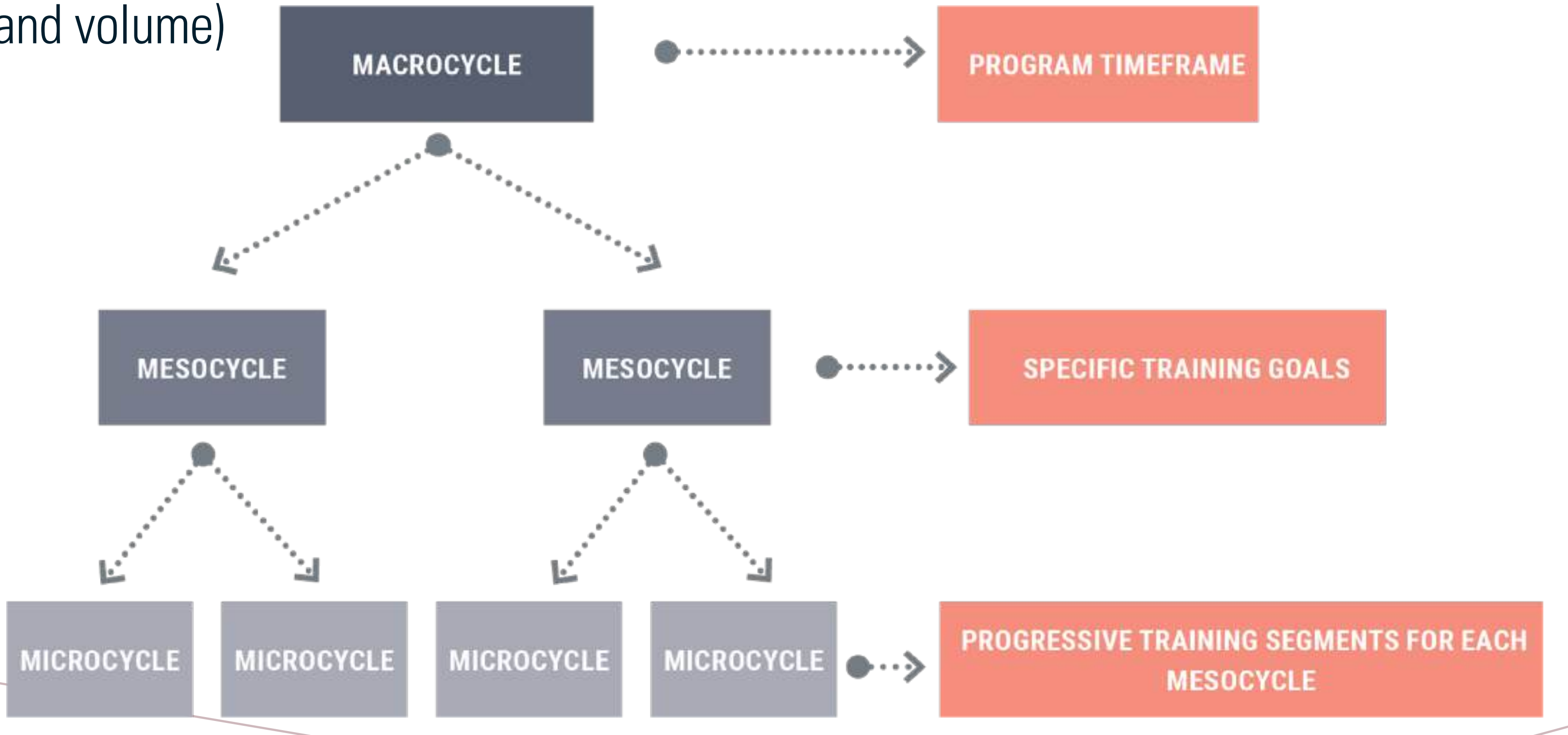


Figure 11-28: General Periodized Program Layout

PERIODISATION

[PG 508]

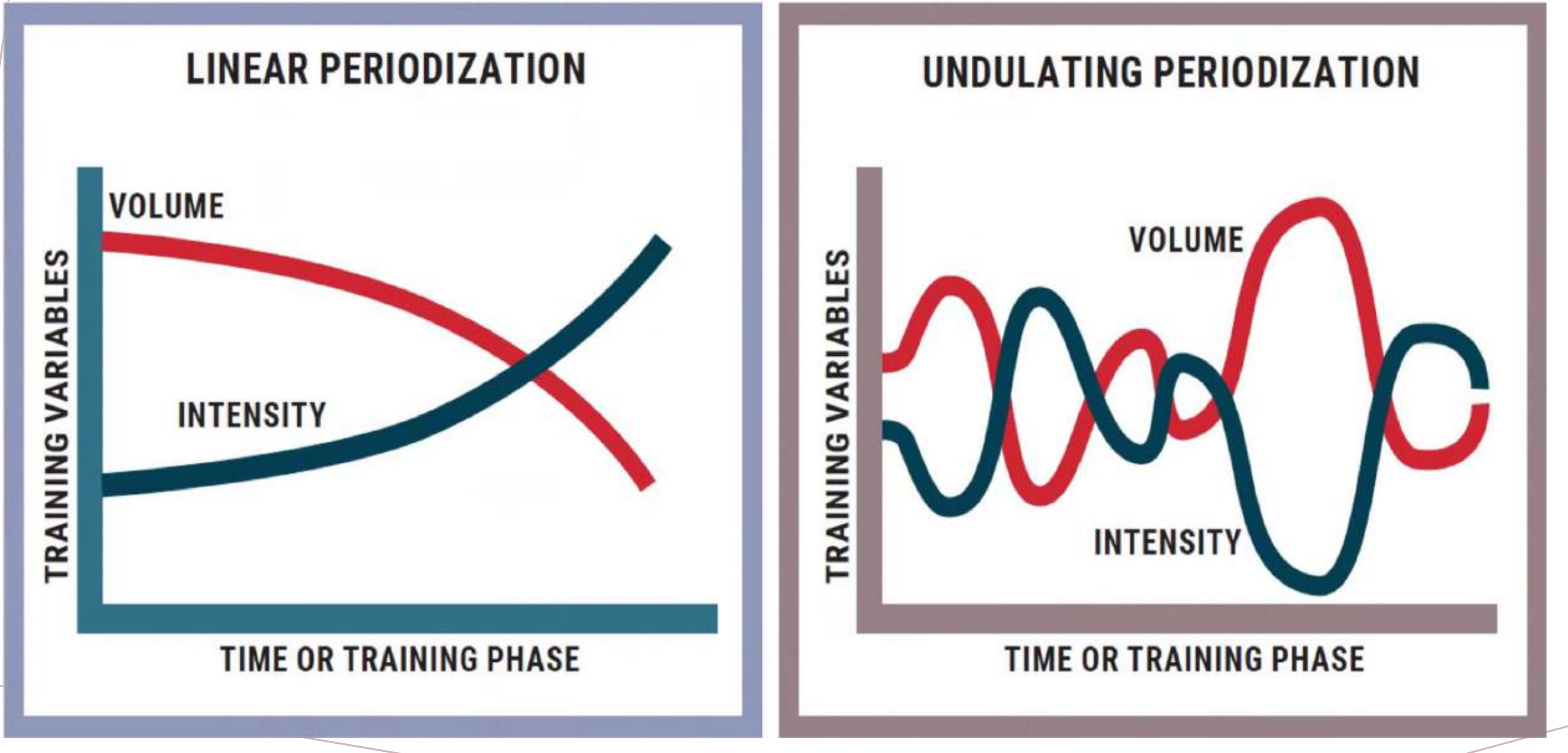


Figure 11-29: Linear vs Undulating Periodisation

PERIODISATION

Table 10-20 Linear Periodization Model (Six Month Macrocycle)

Mesocycle 1 (3 months)	Monday	Wednesday	Friday
Microcycle 1 (2 weeks)	140 lb (64 kg) x 12 repetitions	140 lb (64 kg) x 12 repetitions	140 lb (64 kg) x 12 repetitions
Microcycle 2 (2 weeks)	160 lb (73 kg) x 8 repetitions	160 lb (73 kg) x 8 repetitions	160 lb (73 kg) x 8 repetitions
Microcycle 3 (2 weeks)	180 lb (82 kg) x 4 repetitions	180 lb (82 kg) x 4 repetitions	180 lb (82 kg) x 4 repetitions
Microcycle 4 (2 weeks)	155 lb (70 kg) x 12 repetitions	155 lb (70 kg) x 12 repetitions	155 lb (70 kg) x 12 repetitions
Microcycle 5 (2 weeks)	175 lb (79 kg) x 8 repetitions	175 lb (79 kg) x 8 repetitions	175 lb (79 kg) x 8 repetitions
Microcycle 6 (2 weeks)	195 lb (89 kg) x 4 repetitions	195 lb (89 kg) x 4 repetitions	195 lb (89 kg) x 4 repetitions
Interim Week	230 lb (104 kg) x 1 repetition (goal assessment)	Rest	Rest

Mesocycle 2 (3 months)	Monday	Wednesday	Friday
Microcycle 1 (2 weeks)	170 lb (77 kg) x 12 repetitions	170 lb (77 kg) x 12 repetitions	170 lb (77 kg) x 12 repetitions
Microcycle 2 (2 weeks)	190 lb (86 kg) x 8 repetitions	190 lb (86 kg) x 8 repetitions	190 lb (86 kg) x 8 repetitions
Microcycle 3 (2 weeks)	210 lb (95 kg) x 4 repetitions	210 lb (95 kg) x 4 repetitions	210 lb (95 kg) x 4 repetitions
Microcycle 4 (2 weeks)	180 lb (82 kg) x 12 repetitions	180 lb (82 kg) x 12 repetitions	180 lb (82 kg) x 12 repetitions
Microcycle 5 (2 weeks)	200 lb (91 kg) x 8 repetitions	200 lb (91 kg) x 8 repetitions	200 lb (91 kg) x 8 repetitions
Microcycle 6 (2 weeks)	220 lb (100 kg) x 4 repetitions	220 lb (100 kg) x 4 repetitions	220 lb (100 kg) x 4 repetitions
Interim Week	250 lb (114 kg) x 1 repetition (goal assessment)	Rest	Rest

PERIODISATION

Table 10-21 Undulating Periodization Model (Six Month Macrocycle)

Mesocycle 1 (3 months)	Monday	Wednesday	Friday
Microcycle 1 (2 weeks)	140 lb (64 kg) x 12 repetitions	160 lb (73 kg) x 8 repetitions	180 lb (82 kg) x 4 repetitions
Microcycle 2 (2 weeks)	140 lb (64 kg) x 12 repetitions	160 lb (73 kg) x 8 repetitions	180 lb (82 kg) x 4 repetitions
Microcycle 3 (2 weeks)	140 lb (64 kg) x 12 repetitions	160 lb (73 kg) x 8 repetitions	180 lb (82 kg) x 4 repetitions
Microcycle 4 (2 weeks)	155 lb (70 kg) x 12 repetitions	175 lb (79 kg) x 8 repetitions	195 lb (89 kg) x 4 repetitions
Microcycle 5 (2 weeks)	155 lb (70 kg) x 12 repetitions	175 lb (79 kg) x 8 repetitions	195 lb (89 kg) x 4 repetitions
Microcycle 6 (2 weeks)	155 lb (70 kg) x 12 repetitions	175 lb (79 kg) x 8 repetitions	195 lb (89 kg) x 4 repetitions
Interim Week	230 lb (104 kg) x 1 repetition (goal assessment)	Rest	Rest

Mesocycle 2 (3 months)	Monday	Wednesday	Friday
Microcycle 1 (2 weeks)	170 lb (77 kg) x 12 repetitions	190 lb (86 kg) x 8 repetitions	210 lb (95 kg) x 4 repetitions
Microcycle 2 (2 weeks)	170 lb (77 kg) x 12 repetitions	190 lb (86 kg) x 8 repetitions	210 lb (95 kg) x 4 repetitions
Microcycle 3 (2 weeks)	170 lb (77 kg) x 12 repetitions	190 lb (86 kg) x 8 repetitions	210 lb (95 kg) x 4 repetitions
Microcycle 4 (2 weeks)	180 lb (82 kg) x 12 repetitions	200 lb (91 kg) x 8 repetitions	220 lb (100 kg) x 4 repetitions
Microcycle 5 (2 weeks)	180 lb (82 kg) x 12 repetitions	200 lb (91 kg) x 8 repetitions	220 lb (100 kg) x 4 repetitions
Microcycle 6 (2 weeks)	180 lb (82 kg) x 12 repetitions	200 lb (91 kg) x 8 repetitions	220 lb (100 kg) x 4 repetitions
Interim Week	250 lb (114 kg) x 1 repetition (goal assessment)	Rest	Rest

PROGRAM MAINTENANCE

INTERNATIONAL SPORTS ACADEMY

[PG 511 - 513]

- Following pointers to avoid detraining and maintain cardiometabolic health and fitness
  - The minimum dose of activity to maintain weight and cardiometabolic health equates to 1 mile+ per day (8 mile (12.9 km) /week)
  - Low-dose HIIT (12 minutes/week) improves cardiometabolic health
  - Integrate a single long training session into the biweekly training routine
  - Combined cardiorespiratory and muscular training is an antidote to detraining

RECOVERY

[PG 513]

Figure 11-30: FITT for training recovery

SUMMARY

[PG 477]

Functional Training	▸ Focus on establishing/reestablishing postural stability and kinetic chain mobility. ▸ Exercise programs should improve muscular endurance, flexibility, core function, and static and dynamic balance. ▸ Progress exercise volume and challenge as function improves.
Movement Training	▸ Focus on developing good movement patterns without compromising postural or joint stability. ▸ Programs should include exercises for all five primary movement patterns in varied planes of motion. ▸ Integrate Functional Training exercises to help clients maintain and improve postural stability and kinetic chain mobility.
Load/Speed Training	▸ Focus on application of external loads to movements to create increased force production to meet desired goals. ▸ Integrate the five primary movement patterns through exercises that load them in different planes of motion and combinations. ▸ Integrate Functional Training exercises to enhance postural stability and kinetic chain mobility to support increased workloads. ▸ Programs should focus on adequate resistance training loads to help clients reach muscular strength, endurance, and hypertrophy goals. ▸ Clients with goals for athletic performance will integrate exercises and drills to build speed, agility, quickness, and power.

Table 11-5: Muscular Training